

WEEK 4

CYBER SECURITY INTERNSHIP

Basic Network Setup & Initial Security Testing

SUBMITTED BY	Akash Sanuria
INTERN ID	DG/SEPTEMBER/CYBER/171
ORGANIZATION	DG Interns Hub
INTERNSHIP	Cybersecurity Internship
SUBMISSION	30 September 2026

PROJECT OBJECTIVE

Understand basic network setup and perform initial security testing in an isolated virtual lab environment.

01 Project Overview

This Week 4 mini project demonstrates an end-to-end security lab workflow: creating a virtual network, configuring a target web service, discovering exposed services with Nmap, and analyzing live traffic with Wireshark. Kali Linux was used as the attacker/testing machine and Ubuntu Linux as the target. All activity was performed inside an authorized VirtualBox lab.

Objectives

- Create a small VirtualBox NAT Network and connect Kali Linux with Ubuntu.
- Verify host-to-host communication on the same subnet.
- Install and validate an Apache2 web server on Ubuntu.
- Use Nmap to identify open ports, services, and service versions.
- Capture and analyze ICMP, DNS, and HTTP traffic using Wireshark.
- Save practical evidence and document the workflow with real screenshots.

Tools & Roles

TOOL	ROLE IN THE LAB
Oracle VirtualBox	Virtual machine and network environment
Kali Linux	Attacker / testing machine
Ubuntu Linux	Target machine
Apache2	HTTP service on TCP port 80
Nmap	Port and service discovery
Wireshark	Packet capture and protocol analysis

FINAL ACTIVE LAB ADDRESSES

Kali Linux: 10.0.2.15 | Ubuntu Linux: 10.0.2.5 | Subnet: 10.0.2.0/24. Earlier screenshots may show different addresses because DHCP reassigned the VMs during the practical.

02 Task 1 - Basic Network Lab Setup

A NAT Network named **NatNetwork** was created in VirtualBox with IPv4 prefix **10.0.2.0/24** and DHCP enabled. Both virtual machines were attached to the same virtual network.

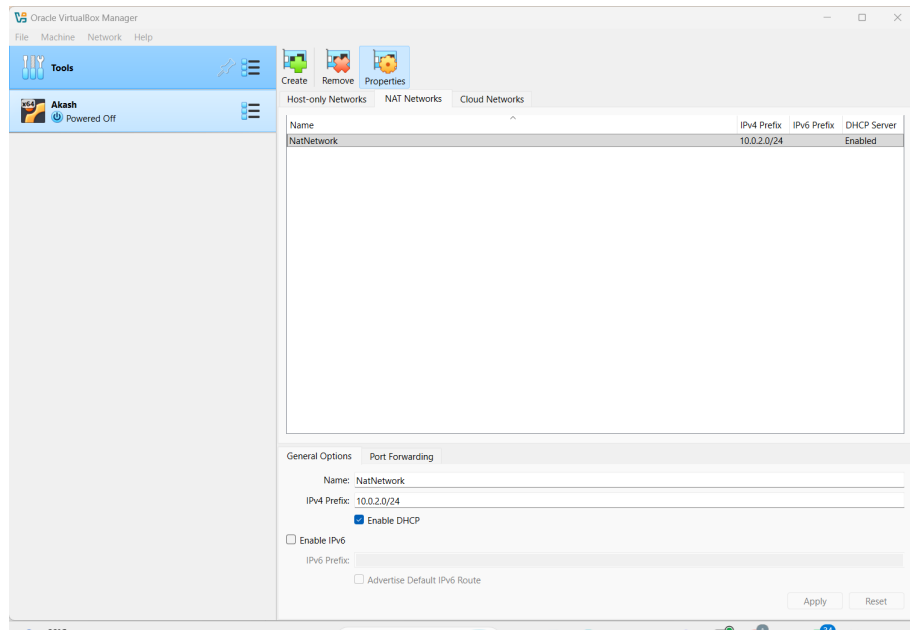


Figure 1. VirtualBox NAT Network with DHCP enabled.

Connectivity Verification

Initial ping testing confirmed that the two virtual machines could communicate. During the practical, DHCP later reassigned the VM addresses; the final working pair was Kali 10.0.2.15 and Ubuntu 10.0.2.5.

```
(akash@akash) [ ]
$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.93 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=2.19 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.790 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=4.13 ms

— 10.0.2.15 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3031ms
rtt min/avg/max/mdev = 0.790/2.260/4.132/1.202 ms
```

Figure 2. Successful Kali-to-Ubuntu connectivity test with 0% packet loss.

RESULT

Both machines were on the same virtual subnet and communication was verified successfully.

03 Task 2 - Apache Web Server

Apache2 was installed on the Ubuntu target using APT. A temporary DNS resolution issue appeared during package updates; direct IP connectivity was working, so DNS was restored before installation continued. The Apache service was then verified with systemctl.

```

vboxuser@Ubuntu: ~
sr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.
service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6)...
Processing triggers for man-db (2.12.0-4build2)...
Processing triggers for libc-bin (2.39-0ubuntu8.9)...
vboxuser@Ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset:
   Active: active (running) since Wed 2026-09-30 13:50:41 UTC; 4min 19s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3602 (apache2)
    Tasks: 55 (limit: 1432)
   Memory: 5.3M (peak: 5.4M)
      CPU: 89ms
   CGroup: /system.slice/apache2.service
           └─3602 /usr/sbin/apache2 -k start
             └─3604 /usr/sbin/apache2 -k start
               └─3605 /usr/sbin/apache2 -k start

Sep 30 13:50:41 Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP S
Sep 30 13:50:41 Ubuntu apachectl[3601]: AH00558: apache2: Could not reliably de
Sep 30 13:50:41 Ubuntu systemd[1]: Started apache2.service - The Apache HTTP S
lines 1-16/16 (END)

```

Figure 3. Apache2 service verified as active (running) on Ubuntu.

From Kali Linux, the Ubuntu target was opened over HTTP. The Apache2 default page confirmed that the web service was reachable across the lab network.

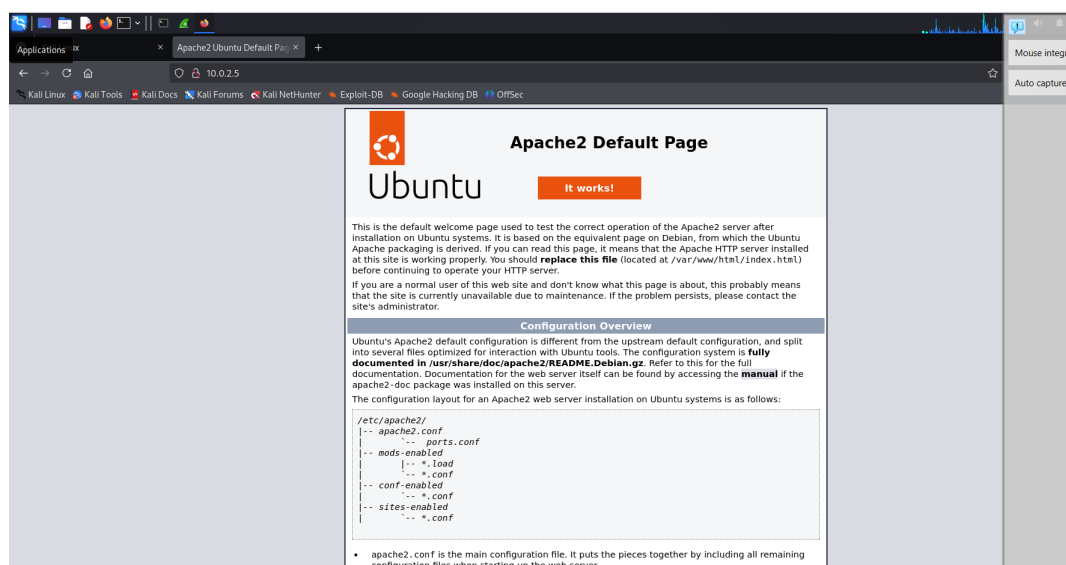


Figure 4. Apache2 default page successfully accessed from Kali Linux.

FINDING

The target exposed an HTTP service on TCP port 80, creating a clear service for Nmap discovery and HTTP packet analysis.

04 Task 3 - Nmap Network Scanning

Service/version detection was performed from Kali against the final Ubuntu target address using **nmap -sV 10.0.2.5**. The -sV option requests service and version identification.

```
(akash@Akash) [~]$ nmap -sV 10.0.2.5 -oN week4_final_nmap.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-30 19:30 IST
Nmap scan report for 10.0.2.5
Host is up (0.00017s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:0F:22:1A (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.95 seconds
```

Figure 5. Final Nmap service/version scan of Ubuntu 10.0.2.5.

PORT	STATE	SERVICE	DETECTED VERSION
80/tcp	Open	HTTP	Apache httpd 2.4.58 (Ubuntu)

```
(akash@Akash) [~]$ cat week4_final_nmap.txt
# Nmap 7.94SVN scan initiated Wed Sep 30 19:30:49 2026 as: /usr/lib/nmap/nmap
--privileged -sV -oN week4_final_nmap.txt 10.0.2.5
Nmap scan report for 10.0.2.5
Host is up (0.00017s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:0F:22:1A (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Wed Sep 30 19:30:56 2026 -- 1 IP address (1 host up) scanned in 6.95 seconds
```

Figure 6. Final Nmap output saved and verified as week4_final_nmap.txt.

SCAN SUMMARY

Host up. 999 default-scanned TCP ports closed. TCP/80 open and identified as Apache HTTP 2.4.58.

05 Task 4 - Wireshark Traffic Analysis

Wireshark was run on Kali Linux using interface **eth0**. Display filters isolated ICMP, DNS, and HTTP traffic so that each protocol could be examined separately.

ICMP Analysis

A ping from Kali 10.0.2.15 to Ubuntu 10.0.2.5 generated alternating Echo Request and Echo Reply packets. The terminal showed four packets transmitted and four received with 0% packet loss.

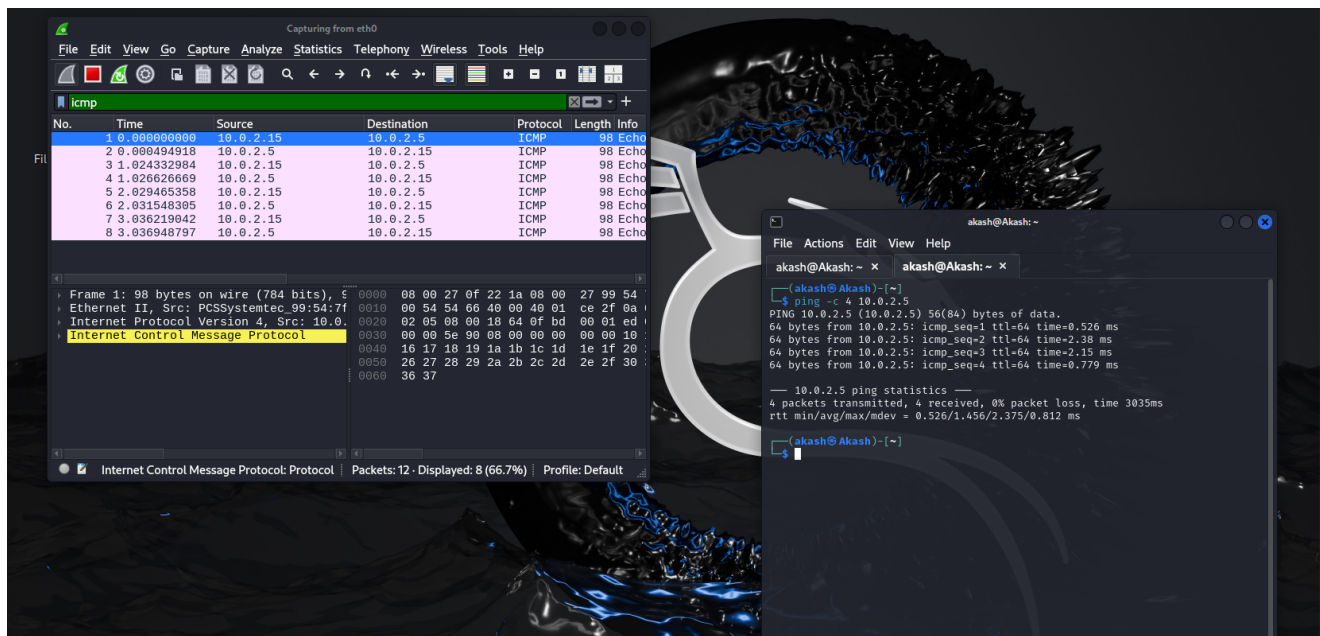


Figure 7. ICMP Echo Request and Echo Reply packets between Kali and Ubuntu.

DNS Analysis

The command **nslookup google.com** generated DNS traffic. The DNS display filter showed query and response packets, while the terminal confirmed successful resolution to multiple IPv4 and IPv6 addresses.

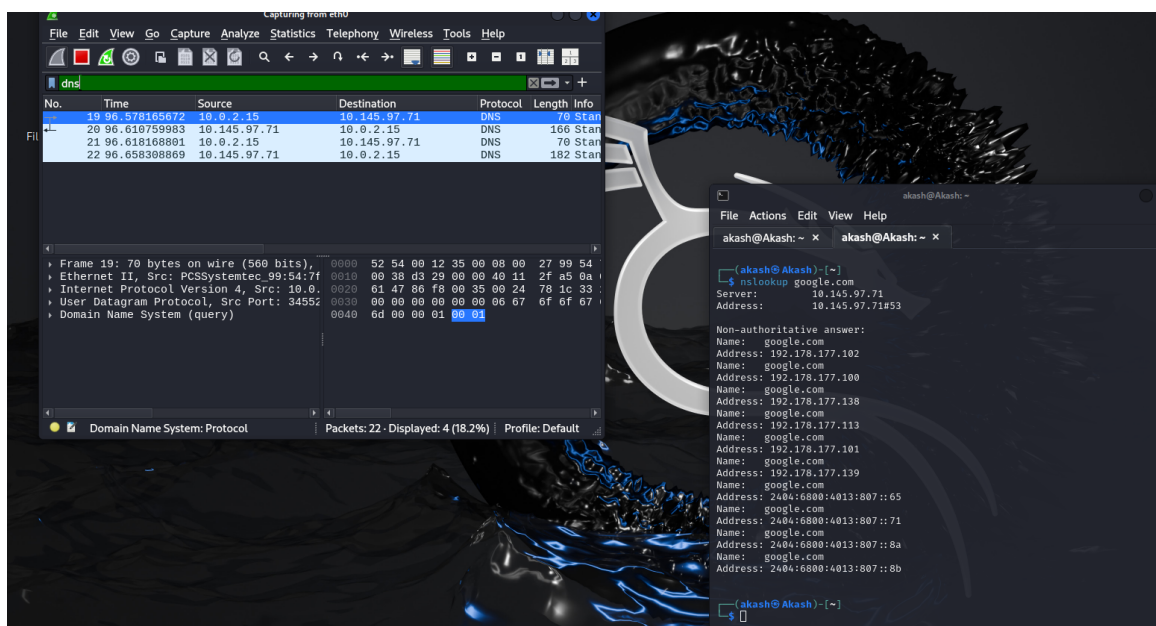


Figure 8. DNS query/response traffic generated by nslookup google.com.

06 HTTP Traffic Analysis

The Apache page at <http://10.0.2.5> was opened from Kali while Wireshark captured traffic. The display filter **http && ip.addr == 10.0.2.5** isolated only HTTP packets involving the Ubuntu target.

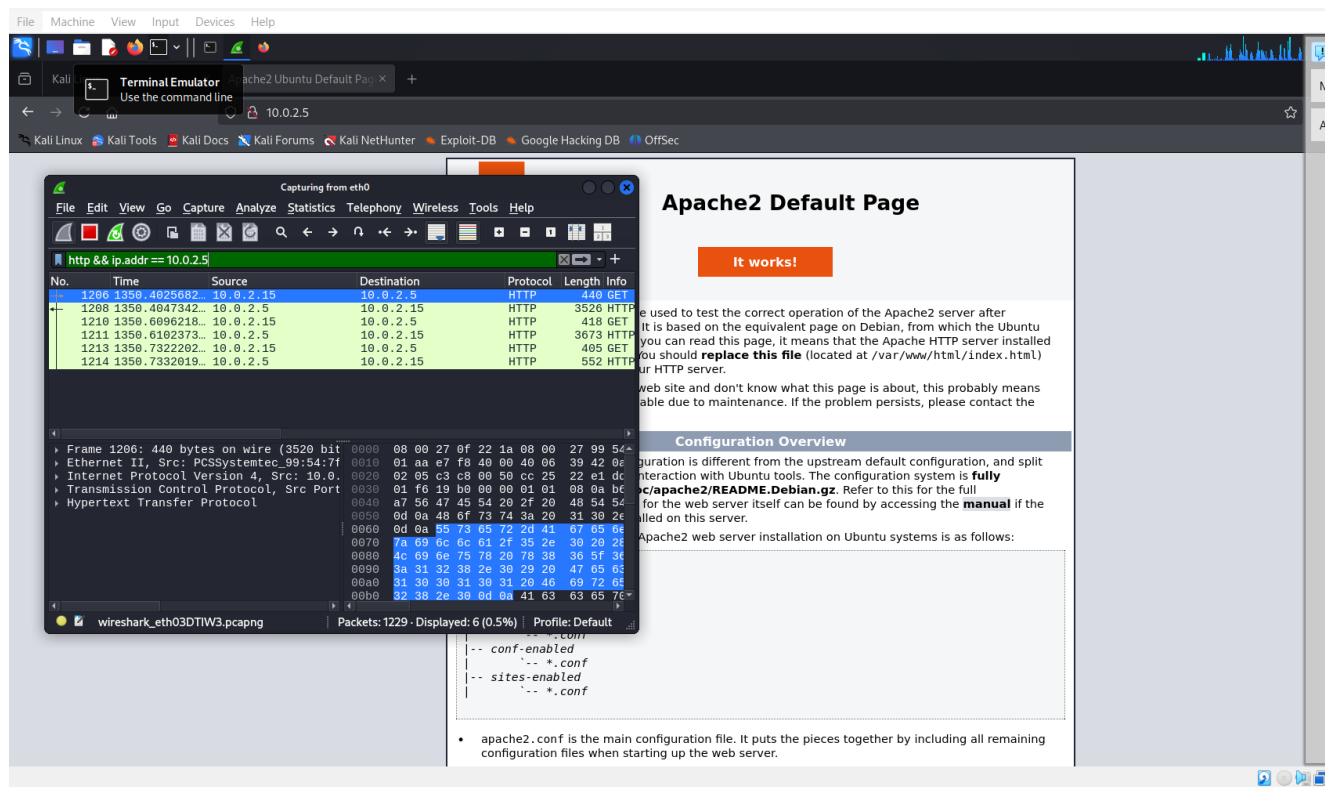


Figure 9. Target-specific HTTP GET requests and responses with the Apache page visible.

SECURITY RELEVANCE

Plain HTTP is not encrypted. Packet analysis can expose protocol metadata and, depending on the application, readable request or response content. HTTPS adds TLS encryption and is preferred for sensitive real-world services.

07 Mini Project Workflow & Key Findings

STEP	ACTIVITY	RESULT
01	Network Setup	Created NatNetwork (10.0.2.0/24) and connected Kali + Ubuntu.
02	Connectivity	Verified host-to-host communication using ICMP ping.
03	Service Setup	Installed Apache2 and verified active (running).
04	Web Access	Opened the Ubuntu Apache page from Kali over HTTP.
05	Scanning	Nmap identified TCP/80 and Apache 2.4.58.
06	Evidence	Saved final scan output to week4_final_nmap.txt.
07	Traffic Analysis	Captured ICMP, DNS, and HTTP traffic with Wireshark.

Key Findings

- Ubuntu was reachable from Kali on the same VirtualBox NAT Network.
- Apache2 exposed TCP port 80 and was accessible from the testing VM.
- Nmap identified the HTTP service and Apache version correctly.
- Wireshark captured successful ICMP request/reply exchanges.
- DNS queries and responses were visible during nslookup testing.
- HTTP GET requests and responses were isolated for the Ubuntu target.
- DHCP address changes after restarts must be checked before scanning.

08 Challenges & Troubleshooting

CHALLENGE	OBSERVATION	RESOLUTION
DNS resolution failure	APT had IP connectivity but could not resolve Ubuntu IP to hostnames	Verified direct connectivity and restored DNS before updating packages
Changing DHCP addresses	After restart, the previous target IP no longer represented the host	Set static IP on Kali and used Kali 10.0.2.15 / Ubuntu 10.0.2.5.
Destination Host Unreachable	Kali attempted to reach an outdated target address and IP resolution failed	Compiling IP resolution table from neighbour data, restarted cleanly, and verified
Apache unavailable	The active Ubuntu session did not have apache2 service installed	Installed Apache2 in the active Ubuntu system and verified active
HTTPS vs HTTP	Firefox initially attempted HTTPS while the lab service was HTTP only	Used Wireshark to filter on TCP/80 and captured the resulting traffic.

Learning Outcomes

- Improved understanding of VirtualBox networking and DHCP-assigned addresses.
- Learned to validate connectivity before running security scans.
- Practiced installing and verifying a Linux web service.
- Learned Nmap service/version scanning and output saving.
- Practiced Wireshark filters for ICMP, DNS, and HTTP.
- Developed a systematic troubleshooting approach.

09 Conclusion

The Week 4 cybersecurity internship task was completed successfully in an authorized virtual lab. A small network was created with Kali Linux and Ubuntu, communication was verified, and an Apache2 web server was configured on the target. Nmap identified the HTTP service on TCP port 80 and the scan output was saved for submission. Wireshark was used to analyze ICMP, DNS, and HTTP traffic and demonstrate how common protocols appear at packet level.

The practical also provided troubleshooting experience with DNS resolution, DHCP address changes, ARP/neighbour state, and service availability. Overall, the project strengthened foundational skills for network security and entry-level SOC/network analysis work.

WEEK 4 STATUS

All five practical tasks completed successfully with real lab evidence.

Final Submission Checklist

STATUS	DELIVERABLE
COMPLETED	Project report with real practical screenshots
COMPLETED	Nmap scan output saved as week4_final_nmap.txt
COMPLETED	Wireshark evidence for ICMP, DNS, and HTTP
COMPLETED	Week 4 presentation prepared from the same lab evidence